Assignment 3 Report

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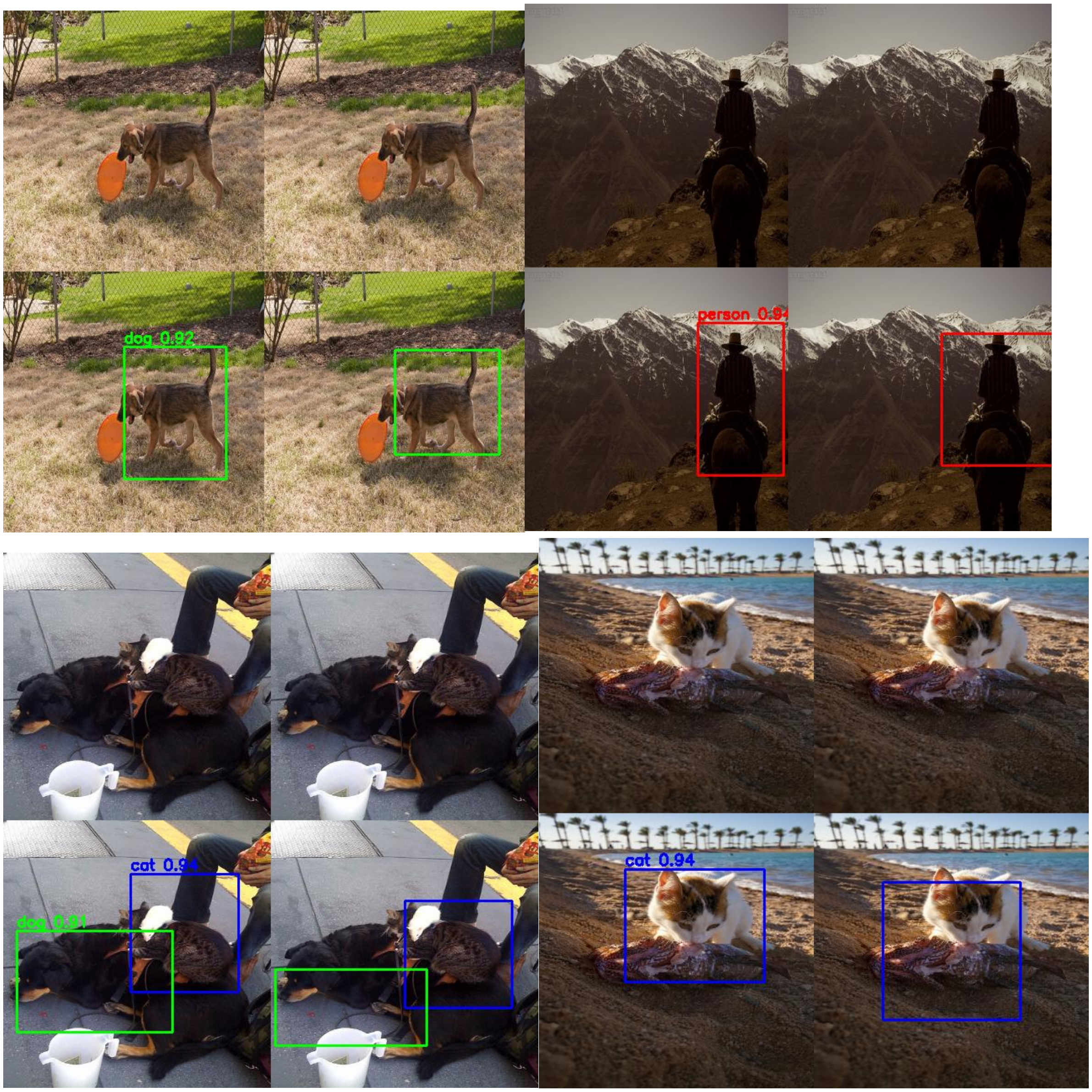
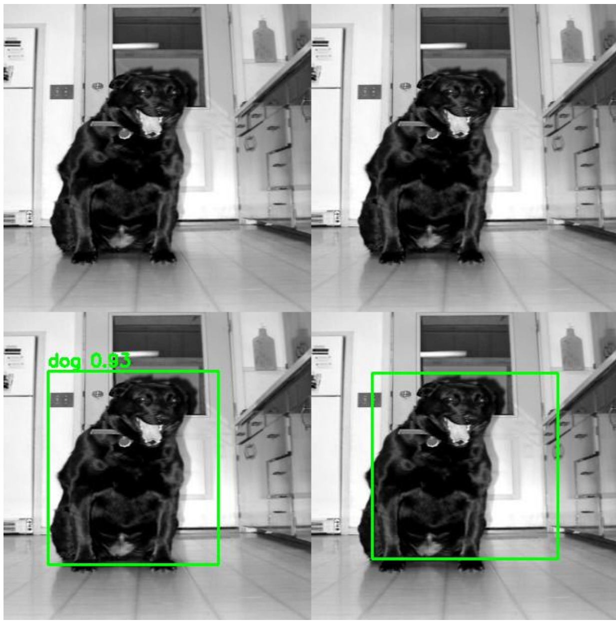
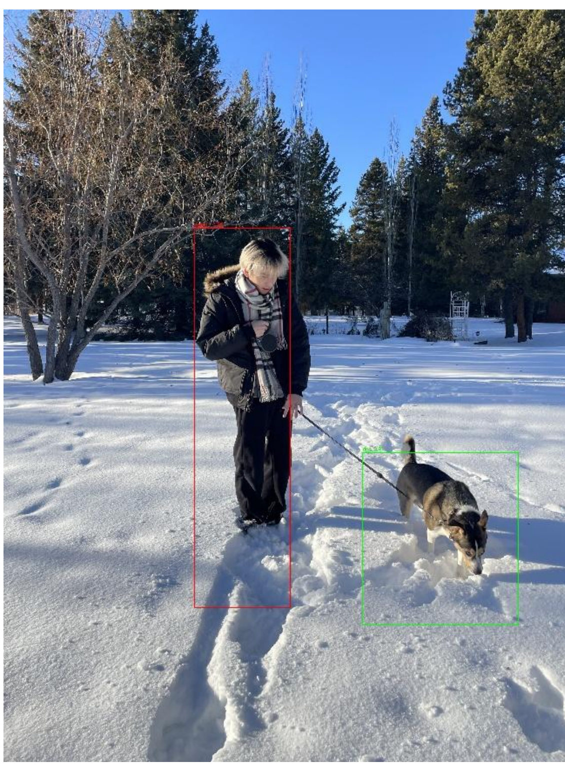
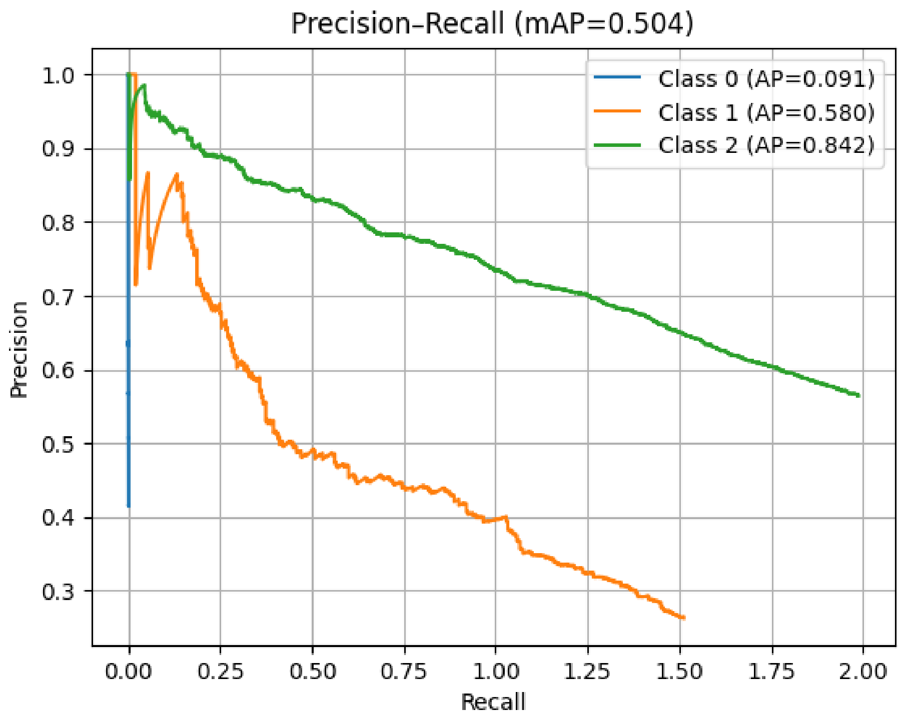

dog 0.92
person 0.94
cat 0.94
dog 0.91
cat 0.94


dog 0.93




FG



Cat AP:0.091

Dog AP:0.580

Person AP:0.842

mAP:0.504

I applied these data argumentations:

HorizontalFlip(p=0.5),

RandomBrightnessContrast(p=0.3),

BBoxSafeRandomCrop(p=0.3),

ColorJitter(p=0.3),

HueSaturationValue(p=0.3),

RandomGamma(gamma_limit=(80, 120), p=0.3)

Why the Model Performs Well

1.Clear Object Boundaries

Objects are well-separated from the background and have a distinct shape. SSD relies heavily on edge contrast + texture cues, so clear silhouettes make detection much easier. The cat and dog in the image lie next to each other but do not overlap heavily. This reduces IoU conflicts during NMS and avoids merging two objects into one false box.

2.Centered & Relatively Large Object

SSD performs best when the object occupies a moderate–large region of the image.
In these images, the object is neither too small nor too close to image borders, so the default boxes align well with its location.

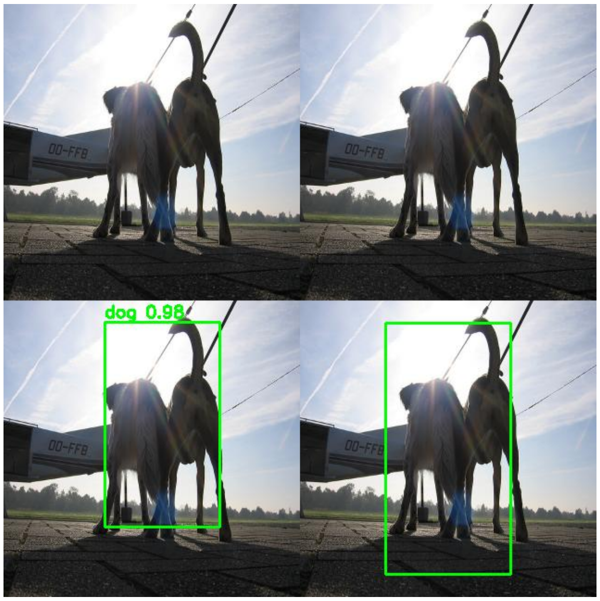
3.Distinct Features (Eyes, Face, Ears)

The object standing pose and the person are similar to common patterns found in the dataset. SSD generalizes easily to these familiar visual structures. Cat faces have extremely strong local features that match the convolution filters learned during training. Even though some backgrounds are complex (beach, concrete floor), the face dominates the region.

4.Good Lighting & Low Occlusion

Lighting is consistent, and the objects are not heavily occluded.
High-visibility regions help the model activate the correct default boxes.

- Cat bounding box is accurate and centered.
- Dog detection is also correct with moderate confidence.
- Predictions remain stable even across varied lighting and textures.



Why the Model Performs Poorly on This Image: can't distinguish 2 dogs facing backwards

1. Extreme Backlighting (Strong Light Source Behind the Dogs)

The sun is directly behind the animals, causing heavy shadows, loss of texture and washed-out edges. SSD relies on clear contours, this lighting makes object boundaries almost invisible.

2. Multiple Animals Overlapping

There are *two dogs standing very close together*, merging into one silhouette.
SSD cannot reliably separate objects that have no clear boundary and appear as one blob.

3. Unusual Viewpoint (Rear View)

Most training data shows side or front views of dogs, not a back view with identical legs overlapping. The model cannot map this unusual shape to a typical "dog" appearance.

What can be done to improve the result?

To improve detection on this exact photo, the model needs more training exposure to strong backlight, silhouettes, overlapping animals, and unusual rear-view poses, plus improved anchor design and a slightly lower threshold.

Why the Model Performs Poorly on This Image: missing one right cat

1. Heavy Occlusion (Person obstructing cats)

The woman’s arm and body block a large portion of the cats.
SSD anchors rely on visible edges and shapes, but most of the cat’s face/body is hidden.

In this image: Two cats overlap on the bottom and the person overlaps both cats. Default boxes fire simultaneously for person and cat classes. NMS cannot suppress all incorrect boxes because boxes belong to different classes SSD is weak when multiple objects overlap or sit extremely close.

2. Cats Have Uncommon Poses + Extreme Foreshortening

The cat on the left is: Sitting with legs hidden. Turned sideways. All white.

The cat on the right: Has its back to the camera. Partially inside a shirt. Very little visible head shape

Which result in low contrast + unfamiliar pose = hard to detect.

3. Indoor Lighting + Warm Wooden Floor → Feature Confusion

The warm floor and cats both share similar **color hues** and **textures**.
Low contrast between object and background reduces SSD’s ability to group correct anchor boxes.

What can be done to improve the results?

Add training images with occluded animals
Include examples where cats/dogs are partially hidden or interacting with people.

Increase dataset diversity
Add indoor images, warm lighting, wooden floor textures, and long-haired cat breeds.

Use stronger augmentation: Random brightness/contrast. Random occlusion. Random cropping

Use a larger SSD model or add more training epochs
More capacity or longer training helps the model learn cat shapes under occlusion.

Why the Model Performs Poorly on This Image: missing dog

1. Strong Occlusion Between Person and Dog

The person is bending forward and covering a large part of the dog.
SSD cannot cleanly separate their shapes, so it extends one big bounding box over both.

2. The Dog Is Small Relative to the Person

Small objects near large objects are difficult for SSD because most default boxes are sized for medium objects. The model "ignores" the tiny dog and focuses on the bigger person.

3. Unusual Pose

The person is kneeling and leaning forward — a pose not common in the training set.
This causes the model to guess an oversized, vertically stretched person box.

4. Low Visual Separation

The dog is white and blends into the bright grass, while the person's dark clothing also merges with the background.
Weak contrast makes edges harder to detect.

What can be done to improve the results?

Add small-object samples into training
Include more images of small dogs relative to the frame size.

Adjust anchor/default box scales
Add anchors for smaller objects so the network has proper priors for tiny animals.

Train with zoom-in / random-crop augmentation
Forces the model to learn small objects better.

What is the best confidence threshold for your your model for the 10 images you chose. You can get this value by running the model at different thresholds and judging the predictions visually.

The best threshold for this is 0.8